

Bézout's Theorem — Proof

Two regular (smooth) complex projective plane curves of degrees n and m , without common components, intersect in exactly nm points, counted with multiplicity.

1 The space of transverse pairs

Let $\mathcal{C}_d = \mathbb{P}^{\binom{d+2}{2}-1}$ be the projective space of all homogeneous polynomials of degree d in three variables, modulo scaling. Let $\Delta_d \subset \mathcal{C}_d$ be the **discriminant locus** — the set of singular curves. Δ_d is a proper Zariski-closed subset (it is the zero locus of the discriminant polynomial in the coefficients). Thus

$$\mathcal{C}_d^{\text{reg}} = \mathcal{C}_d \setminus \Delta_d$$

is a nonempty Zariski-open subset of a projective space. In the ordinary (complex) topology, $\mathcal{C}_d^{\text{reg}}$ is a connected complex manifold: $\mathbb{P}^N \setminus \Sigma$ with Σ a proper analytic subset (real codimension ≥ 2) is connected and path-connected.

A pair $(f, g) \in \mathcal{C}_n^{\text{reg}} \times \mathcal{C}_m^{\text{reg}}$ is **transverse** if at every common point $p \in C_f \cap C_g$, the gradients $\nabla f(p)$ and $\nabla g(p)$ are linearly independent. Define

$$W = \{(f, g) \in \mathcal{C}_n^{\text{reg}} \times \mathcal{C}_m^{\text{reg}} : C_f \pitchfork C_g\}.$$

Lemma 1.1. *W is Zariski-open in $\mathcal{C}_n^{\text{reg}} \times \mathcal{C}_m^{\text{reg}}$, hence connected.*

Proof. Consider the incidence set

$$I = \{(p, f, g) \in \mathbb{CP}^2 \times \mathcal{C}_n \times \mathcal{C}_m : f(p) = 0, g(p) = 0, \text{rank}[\nabla f(p), \nabla g(p)] < 2\}.$$

$f(p) = 0$ and $g(p) = 0$ are polynomial equations in the homogeneous coordinates of p and the coefficient vectors of f, g . The rank condition means that all 2×2 minors of the 3×2 matrix $[\nabla f, \nabla g]$ vanish — again polynomial equations. Hence I is a projective variety (a Zariski-closed subset of $\mathbb{CP}^2 \times \mathcal{C}_n \times \mathcal{C}_m$).

Now consider the projection $\pi : I \rightarrow \mathcal{C}_n \times \mathcal{C}_m$ that forgets the point p . Since $\mathbb{CP}^2$ is projective, the projection of a product with a projective variety is a proper morphism. The image of a

closed set under a proper morphism is closed (this is the fundamental theorem of elimination theory). Therefore $\pi(I)$ is Zariski-closed in $\mathcal{C}_n \times \mathcal{C}_m$.

The set $\pi(I)$ consists of *all* pairs (f, g) (smooth or singular) that have a non-transverse intersection at some common point. Restricting to the smooth locus, $\pi(I) \cap (\mathcal{C}_n^{\text{reg}} \times \mathcal{C}_m^{\text{reg}})$ is closed in the subspace topology, and its complement is exactly W . Hence W is Zariski-open in $\mathcal{C}_n^{\text{reg}} \times \mathcal{C}_m^{\text{reg}}$.

W is nonempty: take n distinct lines and m distinct lines in general position, perturb each to a smooth curve; the intersection remains transverse for small perturbations (see Section 2). A nonempty Zariski-open subset of an irreducible complex variety is Zariski-dense and, in the ordinary topology, is the complement of a closed analytic set of real codimension ≥ 2 , hence connected. $\square$

Lemma 1.2. *The cardinality $\#(C_f \cap C_g)$ is constant on W .*

Proof. For $(f, g) \in W$, both C_f and C_g are compact oriented 2-dimensional submanifolds of $\mathbb{CP}^2$ intersecting transversely. Their oriented intersection number (the sum of ± 1 at each intersection point, according to orientation) is invariant under smooth homotopy of the pair of submanifolds — a standard theorem of differential topology. Moreover, for complex submanifolds intersecting transversely, the intersection sign is always $+1$ (complex manifolds carry a canonical orientation, and a transverse intersection of complex submanifolds preserves it). Thus the intersection number equals the unsigned count $\#(C_f \cap C_g)$.

The function $(f, g) \mapsto \#(C_f \cap C_g)$ is locally constant (it is integer-valued and varies continuously in the C^1 topology on the space of pairs). Since W is connected, it is constant on W . $\square$

2 Computing the constant

Pick n lines $\ell_1, \dots, \ell_n$ and m lines $m_1, \dots, m_m$ in general position:

- no three lines are concurrent,
- no ℓ_i coincides with any m_j .

Set

$$f_0 = \ell_1 \ell_2 \cdots \ell_n, \quad g_0 = m_1 m_2 \cdots m_m.$$

Two distinct lines meet exactly once, so

$$C_{f_0} \cap C_{g_0} = \{ \ell_i \cap m_j : 1 \leq i \leq n, 1 \leq j \leq m \},$$

which consists of nm distinct points, each a transverse intersection (two distinct lines cross normally). However, C_{f_0} and C_{g_0} are not smooth curves: they are unions of lines, singular

at the $\binom{n}{2}$ (resp. $\binom{m}{2}$) points where the component lines cross. We perturb to obtain smooth transverse curves.

Smoothness of small generic perturbations. The discriminant locus $\Delta_d \subset \mathcal{C}_d$ is a proper Zariski-closed subset. Hence $\mathcal{C}_d^{\text{reg}}$ is open and dense. For a fixed $f_0 \in \mathcal{C}_d$, consider the linear family $\{f_0 + tQ : t \in \mathbb{C}\}$ for a given direction $Q \in \mathcal{C}_d$. The condition that $f_0 + tQ \in \Delta_d$ (for some t) is an algebraic condition in the coefficients of Q . For a **generic** choice of Q , the one-parameter family meets Δ_d in at most finitely many values of t (or avoids it entirely for small $t \neq 0$). Consequently, for a generic Q , the curve $C_{f_0 + \varepsilon Q}$ is smooth for all sufficiently small $\varepsilon \neq 0$.

Choose generic degree- n polynomial Q and generic degree- m polynomial R , and set

$$f_\varepsilon = f_0 + \varepsilon Q, \quad g_\delta = g_0 + \delta R.$$

We claim: for all sufficiently small $|\varepsilon|, |\delta| > 0$, $(f_\varepsilon, g_\delta) \in W$ and $\#(C_{f_\varepsilon} \cap C_{g_\delta}) = nm$.

Local analysis near each p_{ij} . Let $p_{ij} = \ell_i \cap m_j$. Choose affine coordinates (x, y) centred at p_{ij} such that $\ell_i = \{y = 0\}$ and $m_j = \{x = 0\}$ (always possible since the two lines are distinct). Because no other ℓ_k or m_k passes through p_{ij} (general position: three lines never concurrent), we can write, near $(0, 0)$,

$$f_0(x, y) = y u(x, y), \quad g_0(x, y) = x v(x, y)$$

with holomorphic u, v satisfying $u(0, 0) \neq 0$, $v(0, 0) \neq 0$. Indeed, $\ell_i = \{y = 0\}$ is a simple factor of f_0 at p_{ij} , and the product of the remaining $n-1$ lines gives a function u nonvanishing at p_{ij} .

The perturbed equations are

$$\begin{cases} f_\varepsilon(x, y) = y u(x, y) + \varepsilon Q(x, y) = 0, \\ g_\delta(x, y) = x v(x, y) + \delta R(x, y) = 0. \end{cases}$$

Define the holomorphic map $F : \mathbb{C}^2 \times \mathbb{C}^2 \rightarrow \mathbb{C}^2$ by $F(\varepsilon, \delta, x, y) = (f_\varepsilon(x, y), g_\delta(x, y))$. At $(\varepsilon, \delta, x, y) = (0, 0, 0, 0)$ we have $F(0, 0, 0, 0) = (0, 0)$, and the Jacobian with respect to (x, y) is

$$J_{(x,y)} = \begin{pmatrix} \partial_x f_\varepsilon & \partial_y f_\varepsilon \\ \partial_x g_\delta & \partial_y g_\delta \end{pmatrix}_{(0,0,0,0)} = \begin{pmatrix} \partial_x(yu)(0,0) & \partial_y(yu)(0,0) \\ \partial_x(xv)(0,0) & \partial_y(xv)(0,0) \end{pmatrix} = \begin{pmatrix} 0 & u(0,0) \\ v(0,0) & 0 \end{pmatrix}.$$

(The εQ and δR terms contribute zero derivatives at $\varepsilon = \delta = 0$.) Hence

$$\det J_{(x,y)} = -u(0,0)v(0,0) \neq 0.$$

By the holomorphic implicit function theorem, there exists a neighbourhood U_{ij} of $(0, 0)$ in the (ε, δ) -plane and unique holomorphic functions $x_{ij}(\varepsilon, \delta)$, $y_{ij}(\varepsilon, \delta)$ on U_{ij} satisfying

$$F(\varepsilon, \delta, x_{ij}(\varepsilon, \delta), y_{ij}(\varepsilon, \delta)) = (0, 0), \quad x_{ij}(0, 0) = y_{ij}(0, 0) = 0.$$

Moreover, by continuity of the Jacobian determinant, $\det J_{(x,y)}(\varepsilon, \delta, x_{ij}, y_{ij}) \neq 0$ on a possibly smaller neighbourhood $U'_{ij} \subset U_{ij}$. Within U'_{ij} , the intersection at (x_{ij}, y_{ij}) is transverse (invertible Jacobian $\implies$ linearly independent gradients).

There are nm such points p_{ij} . Since there are only finitely many, the intersection $V = \bigcap_{i,j} U'_{ij}$ is a neighbourhood of $(0,0)$ in $\mathbb{C}^2$. For $(\varepsilon, \delta) \in V \setminus \{(0,0)\}$ with $|\varepsilon|, |\delta|$ sufficiently small, the nm points (x_{ij}, y_{ij}) remain distinct (they are perturbed from distinct p_{ij} and stay in disjoint small neighbourhoods). Each gives a transverse intersection of C_{f_ε} and C_{g_δ} .

No other intersections. Let B_{ij} be small disjoint open balls in $\mathbb{CP}^2$ around each p_{ij} . The set

$$K = \mathbb{CP}^2 \setminus \bigcup_{i,j} B_{ij}$$

is compact. The unperturbed curves C_{f_0} and C_{g_0} have no common points on K , so the continuous function

$$p \longmapsto |f_0(p)| + |g_0(p)|$$

(using any Hermitian metric on the line bundles $\mathcal{O}(n)$ and $\mathcal{O}(m)$, and evaluating on the unit sphere in $\mathbb{C}^3$ as a representative) attains a positive minimum $\eta > 0$ on K .

For sufficiently small $|\varepsilon|, |\delta|$, the sup-norms $\|\varepsilon Q\|_K$ and $\|\delta R\|_K$ are both smaller than $\eta/2$. Then on K ,

$$\begin{aligned} |f_\varepsilon(p)| + |g_\delta(p)| &\geq (|f_0(p)| + |g_0(p)|) - |\varepsilon Q(p)| - |\delta R(p)| \\ &> \eta - \frac{\eta}{2} - \frac{\eta}{2} = 0. \end{aligned}$$

Hence C_{f_ε} and C_{g_δ} have no common point on K .

Conclusion of the perturbation. For all sufficiently small $|\varepsilon|, |\delta| > 0$ (with our chosen generic Q, R):

- C_{f_ε} and C_{g_δ} are smooth (generic perturbation),
- they intersect transversely at exactly nm points (one near each p_{ij} , no others).

Therefore $(f_\varepsilon, g_\delta) \in W$ and $\#(C_{f_\varepsilon} \cap C_{g_\delta}) = nm$. By Lemma 2, the constant value on W is nm :

$$\boxed{\forall (f, g) \in W, \quad \#(C_f \cap C_g) = nm}.$$

3 Non-transverse intersections via perturbation

Now let $(f, g) \in \mathcal{C}_n^{\text{reg}} \times \mathcal{C}_m^{\text{reg}}$ be a pair of smooth curves whose intersection is possibly tangential. (Two distinct smooth curves cannot share a common component: a smooth plane curve is

irreducible, and two irreducible curves of possibly different degrees coincide only when $n = m$ and $f = \lambda g$ — excluded by hypothesis.) Let

$$C_f \cap C_g = \{p_1, \dots, p_r\}$$

be their (finitely many) intersection points.

Step 1 — the transverse locus is locally connected. Pick a small Euclidean-open neighbourhood U of (f, g) in $\mathcal{C}_n^{\text{reg}} \times \mathcal{C}_m^{\text{reg}}$. The non-transverse locus $Z = U \setminus W$ is the intersection of U with the proper Zariski-closed set $\pi(I) \cap (\mathcal{C}_n^{\text{reg}} \times \mathcal{C}_m^{\text{reg}})$ constructed in Lemma 1. Hence Z is a proper (complex-)analytic subset of the complex manifold U . The complement of a proper analytic set in a connected complex manifold is connected; therefore

$$W \cap U \text{ is connected.}$$

Moreover, $W \cap U$ is nonempty because W is dense (Lemma 1).

Step 2 — intersections are localized. Choose small disjoint open balls $B_1, \dots, B_r \subset \mathbb{CP}^2$ centred at $p_1, \dots, p_r$. On the compact set

$$K = \mathbb{CP}^2 \setminus \bigcup_{j=1}^r B_j,$$

the curves C_f and C_g have no common point, so — exactly as in Section 2 — the continuous function $p \mapsto |f(p)| + |g(p)|$ attains a positive minimum $\eta > 0$ on K . By shrinking U if necessary, we ensure that for every $(f', g') \in U$ the perturbations $f' - f$ and $g' - g$ are so small in sup-norm on K that

$$|f'(p)| + |g'(p)| > 0 \quad \text{for all } p \in K.$$

Consequently, for **all** $(f', g') \in W \cap U$,

$$C_{f'} \cap C_{g'} \subset \bigcup_{j=1}^r B_j.$$

Step 3 — counting in each ball is constant. For $(f', g') \in W \cap U$, every intersection is transverse, and by Lemma 2 (together with the global connectedness of W) we have

$$\#(C_{f'} \cap C_{g'}) = nm.$$

All nm points lie in $\bigcup B_j$. For each $j = 1, \dots, r$ define

$$n_j(f', g') = \#(C_{f'} \cap C_{g'} \cap B_j),$$

the number of transverse intersection points inside B_j .

The function $(f', g') \mapsto n_j(f', g')$ is **continuous** on $W \cap U$: each individual intersection point moves holomorphically with the coefficients (by the implicit function theorem applied at a transverse intersection), and no point can enter or leave B_j without crossing the boundary ∂B_j , which contains no intersections of *any* pair in $W \cap U$ (after the shrinking in Step 2, $|f'| + |g'| > 0$ on K , and $\partial B_j \subset K$).

An integer-valued continuous function on a connected space is constant. Since $W \cap U$ is connected, $n_j(f', g')$ takes the same value for every $(f', g') \in W \cap U$. Denote this constant by

$$\mu_j(f, g) := n_j(f', g') \quad (\text{independent of the choice of } (f', g') \in W \cap U).$$

Thus **each intersection point p_j attracts exactly μ_j transverse intersection points under a generic small perturbation**, and this number depends only on the local geometry of f and g near p_j , not on the particular perturbation chosen.

Step 4 — conclusion. The integer μ_j is the **multiplicity** of the intersection at p_j . Summing over all j ,

$$\sum_{j=1}^r \mu_j(f, g) = \sum_{j=1}^r n_j(f', g') = \#(C_{f'} \cap C_{g'}) = nm,$$

where (f', g') is *any* transverse pair in $W \cap U$. Hence

$$\boxed{\sum_{p \in C_f \cap C_g} \mu_p(f, g) = nm}.$$

- Remarks.**
1. For $(f, g) \in W$ one trivially recovers $\mu_p = 1$ for each of the nm points.
 2. The extension from smooth curves to *arbitrary* curves follows by approximating any curve by smooth ones and applying a further limiting argument; the total multiplicity remains nm .